Section	Ask	Points	What good looks like	What average looks like	What poor looks like	Weightage
		60	80-100%	60-80%	<60%	100.00%
Exploratory Data Analysis	- Problem definition, questions to be answered - Data background and contents - Answer all the key questions asked in this section - Univariate analysis - Bivariate analysis -- Key meaningful observations for each of the plot	10	1) Definition of problem (as per given problem statement with additional views) [1] 2) Observations on the shape of data, data types of various attributes, missing values, and statistical summary [1] 3) Correct answers to all the key questions asked in this section: Univariate: Q1. What is the distribution of mortgage attribute? [0.25] Q2. How many customers have credit cards? [0.25] Bivariate: Q3. What are the attributes that have a strong correlation with the target attribute (personal loan)? [0.5] Q4. How does a customer's interest in purchasing a loan vary with their education? [0.5] Q5. How does a customer's interest in purchasing a loan vary with their age? [0.5] 4) Univariate Analysis (boxplots, histograms, countplots, distribution plots for **all** the variables) [2] 5) Bivariate Analysis - Correlation matrix [1] - Target vs **all** variables [2] 6) Meaningful observations on each of the plot [1] *Marks for key questions to be given based on how many were answered correctly.	1) Definition of problem (as per given problem statements) 2) Observations on data types of various attributes, missing values, statistical summary. 3) Correct answers to some of the key questions in this section (2 Univariate questions and at least 2 Bivariate questions). 4) Univariate Analysis (boxplots, histograms, distribution plots only for **selected** variables) 5) Bivariate Analysis (Correlation matrix, pairplot, target vs **some** variables) *Marks for key questions to be given based on how many were answered correctly.	1) Definition of problem (as per given problem statements) 2) Observations on data types of various attributes, statistical summary. 3) Univariate and Bivariate Analysis (any plot) 4) Correct answers to only a few of the key questions in this section (2 Univariate questions and at least 1 Bivariate question). *Marks for key questions to be given based on how many were answered correctly.	16.67%
Data Preprocessing	- Anomalous Value Detection and Treatment (if needed) - Missing Value Treatment (if needed) - Outlier Detection and Treatment (if needed) - Feature Engineering (if needed) - Data Preparation for Modeling *In case one or more of the above steps are not needed, please share the rationale for the same in your submission	6	1) Treatment of negative values in the Experience column. [1.5] 2) Tried any pre-processing with the ZIP code column - Drop, extract some digits, binning the data with any valid approach and mention a reason behind it. [2] 3) Dropped ID column. [1] 4) Identification of outliers and provided comments on it - Treatment is not necessary, but if treated, a valid reasoning to treat outliers should be provided [1.5]	1) Treatment of negative values in Experience column 2) Dropped ID column. 3) Identification of outlier and provided comments on it - Treatment is not necessary but if treated a valid reasoning to treat outliers should be provided 4) Didnt try to pre-process zip code.	1) Dropped ID column. 2) Didn't treat negative values. 3) Used ZIP code with no proper reasoning. 4) No identification and comments on outliers.	10.00%

Model Building	- Choose the metric of choice with proper rationale - Build the decision tree model and comment on the model performance - Visualize and comment on the decision rules of the model	10	1) Choose the metric of choice with proper rationale [2] 2) Build a decision tree model using Sklearn [3] 3) Comment on model performance across various metrics [2] 4) Visualize and comment on the decision tree/extract the rules from tree. [3]	1) Build a decision tree model using Sklearn 2) Comment on model performance only on metric of choice 3) Visualize the decision tree/extract the rules from tree	1) Build a decision tree model using Sklearn 2) Checked model performance	16.67%
Model Performance Improvement and Final Model Selection	- Try and improve the model performance by pruning (pre-pruning and post-pruning) - Check the performance of the pruned models - Compare the performance of all the models built - Select the final model with proper rationale - Visualize and comment on the decision rules and feature importance for the final model	20	1) Performed pre-pruning by tuning various parameters [6] 2) Perform post-pruning [6] 3) Compare the performance of all the models built and comment on the same [2] 4) Select the final model with valid reasoning [2] 5) Visualize and comment on the decision rules from the final model [2] 6) Checked feature importance and comment on important features of the final model [2]	1) Performed pre-pruning by tuning various parameters 2) Perform post-pruning 3) Select the final model without valid reasoning 4) Visualize the decision rules from the final model	1) Performed pre-pruning or post-pruning 2) Visualize the decision rules from the final model	33.33%
Actionable Insights & Recommendations	- Conclude with key takeaways for the business - what would your advice to the marketing team be regarding campaign management?	6	1) At least 4 actionable insights/conclusions provided [4] 2) At least 2 recommendations mentioned [2] (Recommendations can also include points on additional data sources for further analysis, model implementation in real world, potential business benefits from improving the model, etc.)	1) At least 2 actionable insights/conclusions provided 2) At least 1 recommendation mentioned	1) Some points mentioned which are neither conclusive nor actionable	10.00%

Presentation/Notebook - Overall Quality	- Structure and flow - Crispness - Visual appeal - All key insights and recommendations covered? OR - Structure and flow - Well commented code - All key insights and recommendations covered?	8	1) Clear structure and flow - everything sits well in a story - (Problem - Data overview - Solution overview - Findings - Recommendations) [3] 2) Crispness - not too many words - just enough to keep the focus on key things/points [2] 3) Visual appeal - use of charts, colors, diagrams, format, symmetry - informative visualizations that are easy to interpret [2] 4) All key insights and recommendations covered - all key ones from EDA are stressed upon & important insights are not left just in the notebook. clear points on what business should do with this model and how will that be beneficial [1] (These are key to differentiating a good and an excellent solution) OR 1) Well structured notebook with a logical flow [4] 2) Clean and well commented code [4]	- There is structure and flow but some bits are missing / jumbled - Points are made but in too many words - lesser charts, format is not the cleanest - key insights are covered but some points are missed, like actionable recommendations OR - There is structure and flow but some bits are missing - Some of the code is commented	- No structure or flow - Only a few points are covered - story is not complete - Not many visuals used - Final insights and recommendations are missing OR - No structure or flow - No comments in the code	13.33%